

HOMEWORK 1: CHAPTER 1

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Problems.

1. A busy airport sees 1500 takeoffs per day. Prove that there are two planes that must take off within a minute of each other.

Proof. There are 1440 minutes within a day. Let $n = 1500$ and $k = 1440$. If we distribute k planes into k minutes, then we will have 60 planes leftover ($1500 - 1440 = 60$). By the Pigeonhole Principle, we can conclude that there will be at least two planes taking off within the same minute. $\square$

2. Find all triples of positive integers $a < b < c$ for which

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$$

holds.

Proof. $a, b, c \in \mathbb{N}$. $1 < a < b < c \implies \frac{1}{a} > \frac{1}{b} > \frac{1}{c}$. $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1 \implies ab + ac + bc = abc \implies 3bc \geq abc \implies 3 \geq a$. Let $a = 2$: $\frac{1}{2} + \frac{1}{b} + \frac{1}{c} = 1 \implies \frac{1}{b} + \frac{1}{c} = \frac{1}{2} \implies \frac{2}{b} \geq \frac{1}{2} \rightarrow 4 \geq b$. Let $b = 3$: $\frac{1}{2} + \frac{1}{3} + \frac{1}{c} = 1 \implies c = 6$. Therefore, $a = 2, b = 3, c = 6$. $\square$

6. The set M consists of nine positive integers, none of which has a prime divisor larger than six. Prove that M has two elements whose product is the square of an integer.

Proof. We will use the Pigeonhole Principle to prove this. Let $M = \{2^x \cdot 3^y \cdot 5^z : x, y, z \in \mathbb{N} \text{ (0 included)}\}$. We must show that, no matter what, x, y, z will become even when multiplied with another element in M . We can do this through permutations; the following are the possibilities of x, y, z being even or odd, denoted by an e or o : $(o, o, o), (o, e, o), (o, o, e), (e, o, o), (e, e, e), (e, e, o), (o, e, e), (e, o, e)$. Using Pigeonhole Principle, $9 > 8$. Therefore, by Pigeonhole Principle, there are two integers whose product make a perfect square. $\square$

10. Prove that among 502 positive integers, there are always two integers so that either their sum or their difference is divisible by 1000.

Proof. The list of integers that are divisible by 1000 are $\{0, 1, \dots, 999\}$. The list of numbers when added together become divisible by 1000 are as follows: $(0), (1, 999), \dots, (500)$. This would give us 501 total combinations that are divisible by 1000. We have 502 positive integers. Therefore, we truly only have 500 options to choose from, since 0 isn't positive nor negative. But, by Pigeonhole principle, we see that $502 > 500$. Thus, by Pigeonhole Principle, there is at least one integer that can belong to a combination/pair that we've created. $\square$

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General Form. Claim: Let n, m and r be positive integers so that $n > rm$. Let us distribute n identical balls into m identical boxes. Then, there will be at least one box into which we place at least $r + 1$ balls.

Proof. Let's assume the contrary is true, and say that there isn't one box that has $r + 1$ balls. But this contradicts our objective of distributing n balls. Therefore there must be at least one box containing $r + 1$ boxes. $\square$